

HOMEWORK 11 SOLUTIONS

SECTION 6.4

1c

The standard basis of M_{22} is $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$, so by the Steinitz exchange Lemma there is some matrix in this basis that may be replaced by $\vec{v}$ and still produce a basis of M_{22} .

Let's try

$$B = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

Is B independent?

$$a \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + d \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a & a+b \\ a+c & a+d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow a=0, b=0, c=0, d=0 \quad \checkmark$$

Thus, B is an independent set in M_{22} with 4 elements. Because $\dim M_{22} = 4$, we get

$$B = \left\{ \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ is a basis of } M_{22} \text{ containing } \vec{v} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

2b

Let's try the subset $B = \{x+z, x^2-2x-1, x^2+x\} \subseteq \{x^2+3, x+z, x^2-2x-1, x^2+x\}$.

Is B independent?

$$a(x+z) + b(x^2-2x-1) + c(x^2+x) = 0 \Rightarrow (b+c)x^2 + (a-2b+c)x + (2a-b) = 0$$

$$\Rightarrow \begin{cases} 2a-b = 0 \\ a-2b+c = 0 \\ b+c = 0 \end{cases}$$

$$\Rightarrow a=b=c=0 \quad (\text{Gaussian elimination}).$$

Thus, B is an independent set with 3 elements. Since $\dim P_2 = 3$, we conclude:

$$B = \{x+z, x^2-2x-1, x^2+x\} \text{ is a basis of } P_2.$$

5b

Since $S = \{2x^3, 1+x, 3, 1+x+x^2+x^3\}$ has 4 elements and $\dim P_3 = 4$, we just have to check if S is independent.

$$a(2x^3) + b(1+x) + c \cdot 3 + d \cdot (1+x+x^2+x^3) = 0 \Rightarrow (b+3c+d) + (b+d)x + (2a+d)x^2 + dx^3 = 0$$

$$\Rightarrow \begin{cases} b+3c+d = 0 \\ b+d = 0 \\ 2a+d = 0 \\ d = 0 \end{cases} \Rightarrow a=b=c=d=0$$

Therefore S is independent, so S is a basis of P_3 .

SECTION 7.1

1

g) $T: P_n \rightarrow \mathbb{R}$, $T(r_0 + r_1x + \dots + r_nx^n) = r_n$.

1) Is $T(p+q)(x) = T(p(x)) + T(q(x))$ for each $p(x), q(x) \in P_n$?

$$p(x) = a_0 + a_1x + \dots + a_nx^n$$

$$q(x) = b_0 + b_1x + \dots + b_nx^n$$

$$\left. \begin{matrix} p(x) = a_0 + a_1x + \dots + a_nx^n \\ q(x) = b_0 + b_1x + \dots + b_nx^n \end{matrix} \right\} \Rightarrow (p+q)(x) = (a_0+b_0) + (a_1+b_1)x + \dots + (a_n+b_n)x^n \Rightarrow T((p+q)(x)) = a_n+b_n = T(p(x)) + T(q(x)) \checkmark$$

2) Is $T((ap)(x)) = aT(p(x))$ for all $p(x) \in P_n$ and $a \in \mathbb{R}$?

$$p(x) = a_0 + a_1x + \dots + a_nx^n \Rightarrow (ap)(x) = aa_0 + (aa_1)x + \dots + (aa_n)x^n \Rightarrow T((ap)(x)) = aa_n = aT(p(x)) \checkmark$$

We conclude that T is linear.

f) $T: \mathbb{R}^n \rightarrow V$, $T(r_1, \dots, r_n) = r_1\vec{e}_1 + \dots + r_n\vec{e}_n$.

1) Is $T(\vec{x}+\vec{y}) = T(\vec{x}) + T(\vec{y})$ for all $\vec{x}, \vec{y} \in \mathbb{R}^n$?

$$\vec{x} = (x_1, \dots, x_n)$$

$$\vec{y} = (y_1, \dots, y_n)$$

$$\left. \begin{matrix} \vec{x} = (x_1, \dots, x_n) \\ \vec{y} = (y_1, \dots, y_n) \end{matrix} \right\} \Rightarrow \vec{x} + \vec{y} = (x_1+y_1, \dots, x_n+y_n) \Rightarrow T(\vec{x}+\vec{y}) = (x_1+y_1)\vec{e}_1 + \dots + (x_n+y_n)\vec{e}_n = x_1\vec{e}_1 + \dots + x_n\vec{e}_n + y_1\vec{e}_1 + \dots + y_n\vec{e}_n = T(\vec{x}) + T(\vec{y}) \checkmark$$

2) Is $T(a\vec{x}) = aT(\vec{x})$ for all $\vec{x} \in \mathbb{R}^n$ and $a \in \mathbb{R}$?

$$\vec{x} = (x_1, \dots, x_n) \Rightarrow a\vec{x} = (ax_1, \dots, ax_n) \Rightarrow T(a\vec{x}) = (ax_1)\vec{e}_1 + \dots + (ax_n)\vec{e}_n = a(x_1\vec{e}_1 + \dots + x_n\vec{e}_n) = aT(\vec{x}) \checkmark$$

Thus, T is linear.

4a

Let's write $\vec{v} = (2, 1)$ as a linear combination of $(1, 2)$ and $(-1, 0)$:

$$(2, 1) = a(1, 2) + b(-1, 0) \Rightarrow (2, 1) = (a - b, 2a) \Rightarrow a = \frac{1}{2}, b = -\frac{3}{2}, \text{ so } \vec{v} = \frac{1}{2}(1, 2) - \frac{3}{2}(-1, 0)$$

Since T is linear:

$$T(\vec{v}) = \frac{1}{2}T(1, 2) - \frac{3}{2}T(-1, 0) = \frac{1}{2}(1, 0, 1) - \frac{3}{2}(0, 1, 1) = \left(\frac{1}{2}, -\frac{3}{2}, -1\right).$$

5a

Let's write $\vec{v}$ as a linear combination of $\vec{v} + \vec{w}$ and $2\vec{v} - \vec{w}$:

$$\vec{v} = a(\vec{v} + \vec{w}) + b(2\vec{v} - \vec{w}) \Rightarrow \vec{v} = (a + 2b)\vec{v} + (a - b)\vec{w} \Rightarrow \begin{cases} a + 2b = 1 \\ a - b = 0 \end{cases} \Rightarrow a = b = \frac{1}{3}, \text{ so } \vec{v} = \frac{1}{3}(\vec{v} + \vec{w}) + \frac{1}{3}(2\vec{v} - \vec{w}).$$

Since T is linear,

$$T(\vec{v}) = \frac{1}{3}T(\vec{v} + \vec{w}) + \frac{1}{3}T(2\vec{v} - \vec{w}) = \frac{1}{3}(\vec{v} - 2\vec{w}) + \frac{1}{3}(2\vec{v}) = \vec{v} - \frac{2}{3}\vec{w}$$

We can find $T(\vec{w})$ similarly, but there's a lazy alternative:

$$T(\vec{v} + \vec{w}) = \vec{v} - 2\vec{w} \Rightarrow T(\vec{v}) + T(\vec{w}) = \vec{v} - 2\vec{w} \Rightarrow \vec{v} - \frac{2}{3}\vec{w} + T(\vec{w}) = \vec{v} - 2\vec{w} \Rightarrow T(\vec{w}) = -\frac{4}{3}\vec{w}$$

SECTION 7.2

2

$$b) T: P_2 \rightarrow \mathbb{R}^2, T(p(x)) = (p(0), p(1))$$

• BASIS FOR $\text{Ker } T$

Let $p(x) = a + bx + cx^2 \in P^2$. When is $p(x) \in \text{Ker } T$?

$$p(x) \in \text{Ker } T \Leftrightarrow T(p(x)) = (p(0), p(1)) = (0, 0) \Leftrightarrow (a, a + b + c) = (0, 0) \Leftrightarrow \begin{cases} a = 0 \\ a + b + c = 0 \end{cases}$$

The solutions to the latter system are

$$\begin{cases} a = 0 \\ b = -t \\ c = t \end{cases} \quad (t \in \mathbb{R})$$

so the elements of $\text{Ker } T$ are the polynomials $p(x) = -tx + tx^2 = t(x^2 - x)$, $t \in \mathbb{R}$. This says that $B = \{x^2 - x\}$ is a spanning set for $\text{Ker } T$. It is also independent because $x^2 - x \neq 0$, so

$$B = \{x^2 - x\} \text{ is a basis for } \text{Ker } T$$

• BASIS FOR $\text{im} T$.

Note that, by the dimension theorem, $\dim(\text{Ker} T) + \dim(\text{im} T) = \dim P^2 \Rightarrow \dim(\text{im} T) = 2 = \dim \mathbb{R}^2 \Rightarrow \text{im} T = \mathbb{R}^2$.

So in reality $\text{im} T = \mathbb{R}^2$ (i.e. T is onto), so any basis of $\mathbb{R}^2$ will work. For example:

$C: \{(1,0), (0,1)\}$ is a basis of $\text{im} T$

e) $T: M_{22} \rightarrow M_{22}, T \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a+b & b+c \\ c+d & d+a \end{bmatrix}$

• BASIS FOR $\text{Ker} T$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in M_{22}$. When is $A \in \text{Ker} T$?

$$A \in \text{Ker} T \Leftrightarrow T(A) = \begin{bmatrix} a+b & b+c \\ c+d & d+a \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Leftrightarrow \begin{cases} a+b=0 \\ b+c=0 \\ c+d=0 \\ d+a=0 \end{cases}$$

The solutions to the system on the right are:

$$\begin{cases} a = -t \\ b = t \\ c = -t \\ d = t \end{cases} \quad (t \in \mathbb{R})$$

so the matrices in $\text{Ker} T$ are those of the form $\begin{bmatrix} -t & t \\ -t & t \end{bmatrix} = t \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}, t \in \mathbb{R}$. Consequently, $\left\{ \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \right\}$ spans $\text{Ker} T$. This set is also independent because $\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \neq 0$. Therefore:

$B = \left\{ \begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix} \right\}$ is a basis of $\text{Ker} T$

• BASIS FOR $\text{im} T$

By the dimension theorem, $\dim(\text{Ker} T) + \dim(\text{im} T) = \dim M_{22} \Rightarrow \dim(\text{im} T) = 3$.

Also, since $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is a basis of M_{22} , the set $\left\{ T \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, T \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, T \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, T \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\}$ spans $\text{im} T$. In particular, it is not independent (because $\dim(\text{im} T) = 3$), but it contains a basis of $\text{im} T$.

Let's try $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \right\}$. If this set is independent, then it's automatically a basis of $\text{im} T$.

$$a \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} a+b & b+c \\ c & a \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow a=b=c=0.$$

Thus, this set is independent, so

$$C = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\} \text{ is a basis of } \text{im } T$$

SECTION 7.3

1

a) $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $T(x, y, z) = (x+y, y+z, z+x)$.

Since T is a transformation between spaces of the same dimension, it's enough to check if T is one-to-one, i.e., if $\text{Ker } T = \{\vec{0}\}$.

Take $\vec{x} = (x, y, z) \in \mathbb{R}^3$. Then

$$\vec{x} \in \text{Ker } T \Leftrightarrow T(\vec{x}) = \vec{0} \Leftrightarrow (x+y, y+z, z+x) = (0, 0, 0) \Leftrightarrow \begin{cases} x+y = 0 \\ y+z = 0 \\ x+z = 0 \end{cases} \Leftrightarrow x=y=z=0 \Leftrightarrow \vec{x} = \vec{0}.$$

↑
Gaussian elimination

Thus, $\text{Ker } T = \{\vec{0}\}$, so T is one-to-one and thus an isomorphism.

h) $T: M_{nm} \rightarrow M_{mn}$, $T(A) = A^T$

Since $\dim M_{nm} = \dim M_{mn} = mn$, it's enough to check that T is one-to-one.

Let $A \in M_{nm}$, then

$$A \in \text{Ker } T \Leftrightarrow T(A) = A^T = 0 \Leftrightarrow A = (A^T)^T = 0^T = 0$$

Therefore, $\text{Ker } T = \{0\}$, so T is one-to-one (and thus an isomorphism).

4d

$$ST \begin{bmatrix} a & b \\ c & d \end{bmatrix} = S \begin{bmatrix} c & a \\ d & b \end{bmatrix} = \begin{bmatrix} c & 0 \\ 0 & b \end{bmatrix}, \quad TS \begin{bmatrix} a & b \\ c & d \end{bmatrix} = T \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix} = \begin{bmatrix} 0 & a \\ d & 0 \end{bmatrix}. \text{ In particular, } TS \neq ST.$$

6a

$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $T(x, y, z) = (x+y, y+z, z+x)$.

We saw on 1a that T is an isomorphism, so it has an inverse $T^{-1}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$. Let's compute it.

Let $\vec{x} = (x, y, z) \in \mathbb{R}^3$. Then $T^{-1}(\vec{x})$ is the unique solution to the equation $T(\vec{u}) = \vec{x}$.

If $\vec{u} = (\alpha, \beta, \gamma) \in \mathbb{R}^3$, then $T(\vec{u}) = \vec{x} = (\alpha + \beta, \beta + \gamma, \gamma + \alpha) = (x, y, z)$, so $\vec{u}$ solves the system

$$\begin{cases} \alpha + \beta = x \\ \beta + \gamma = y \\ \alpha + \gamma = z \end{cases}$$

Using Gaussian elimination, we see that

$$\alpha = \frac{x - y + z}{2}, \quad \beta = \frac{x + y - z}{2}, \quad \gamma = \frac{-x + y + z}{2}$$

We conclude that the inverse of T is

$$T^{-1}(x, y, z) = \frac{1}{2}(x - y + z, x + y - z, -x + y + z), \quad (x, y, z) \in \mathbb{R}^3$$